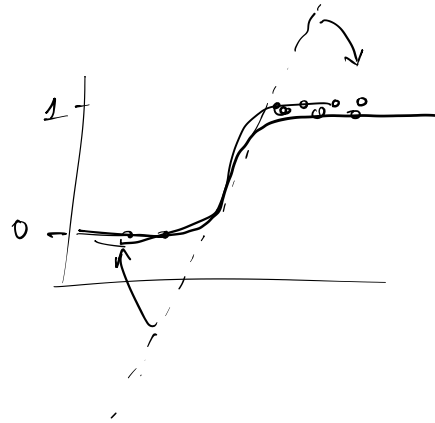
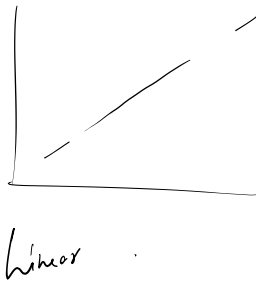


Classification is a supervised ML algorithm/technique used to predict the discrete categories.

Types of Classification Tasks :-

- ① Binary Classification - 2-categories are available in Target
- ② Multi-class Classification - More than 2-categories are available
- ③ Multi-Label Classification - Single data-point has two labels.

① Logistic Regression → Binary Classification Technique



① Model Evaluation

① Confusion Matrix: Compares the actual value & predicted values

Binary Classification
Actual value = 0, 1
Predicted value = 0, 1

	actual	
	1	0
predic	1	True Positive
	0	False +ve
0	1	False -ve
	0	True -ve

$$\text{Accuracy} = \frac{TN + TP}{FP + TN + TP + FN}$$

Case ① Actual = 0, Predicted = 1 → FP

② Actual = 0, Predicted = 0 } TN

③ Actual = 1, Predicted = 1 } TP increase the accuracy

④ Actual = 1, Predicted = 0 } FN